



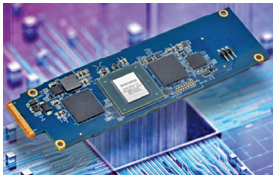
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

GenAI module for edge AI

Unigen Corporation has announced the expansion of its AI portfolio with the introduction of the Amaretti E1.S AI module. The launch marks the company's first hardware platform for generative AI (GenAI), building on its Edge AI lineup. Designed to address the



demand for localised GenAI processing, the Amaretti E1.S delivers a high-density, on-premises solution for AI workloads. The module is designed to power GenAI and large language model applications supporting models with up to 20 billion parameters. It delivers up to 60 trillion operations per second (TOPS) while consuming 10 watts of power. The module is suitable for edge AI applications.

Unigen Corporation
<https://unigen.com>

Wireless vibration sensor for industrial machinery

Crystal Instruments introduces Ruby, a wireless vibration sensor for monitoring industrial machines. Ruby is



designed to help companies detect machine problems before they cause downtime. It measures vibration and shock to help engineers monitor machine condition.

The sensor provides 3-axis vibration measurements and supports Bluetooth and USB connections for accessing data on mobile devices and computers. It also works with RCM software for

downloading recordings and vibration data. Ruby is designed for industrial setups where hundreds of sensors can be connected across machines and production lines using wireless gateways for centralised monitoring. It has IP67 protection against dust and water, operates in temperatures from -40°C to 55°C, and includes a metal housing for industrial use.

Crystal Instruments
<https://www.crystallinstruments.com>

Spectrum sensor for operations at the tactical edge

Silvus Technologies has announced the launch of the FASST 6000 spectrum sensor. Built for operations at the tactical edge, the handheld signal detection



system delivers the performance of larger platforms in a low size, weight and power design. Engineered for mobile signals-intelligence missions and deployable RF sensing networks, the FASST 6000 provides warfighters and unmanned systems operators with situational awareness in contested environments. Delivering scanning performance more than 100 times faster than competing systems in its class, the sensor provides a smaller, lighter, and more affordable solution for dismounted operators and mobile platforms.

Silvus Technologies
<https://silvustechologies.com>

Obstacle-detection chipset for automobiles

Bosch has launched ultrasonic chipset

comprising the TB193 and TB293 that help vehicles detect nearby objects like parked cars, curbs, and obstacles and measure distance. The system captures and processes raw sensor signals



directly to give the vehicle information about its surroundings. The TB293

chip is placed at the sensor to collect signals, while the TB193 chip manages data from multiple sensors and sends it for processing. This setup allows carmakers to adjust the number of sensors based on vehicle models and features and supports software-defined vehicle designs. The system has a data transmission speed of 1.16Mbps, enabling it to process large volumes of data in real time for driver-assistance systems. It also uses up to 50 per cent less power than similar solutions, helping save battery life in hybrid and electric vehicles while reducing wiring, cabling, and component weight by up to 50 per cent.

Bosch
<https://www.bosch-presse.de>

Wireless power supply chipset for wearables

ROHM Semiconductor has introduced a wireless power supply IC chipset, the ML7670 receiver and ML7671 transmitter, designed for NFC-based wireless



charging in wearable devices such as smart rings and smart bands,

as well as accessories like smart pens. The chipset supports up to 250mW of wireless power transfer and integrates peripheral components, including the switching MOSFETs required to power charging ICs, directly into the system, reducing board space and improving power transfer efficiency for compact